Artificial Intelligence
Lecture 11: Probabilistic reasoning

Prof. Dr. Pierre-Alexandre Murena

SoSe 2025

Outline

1 Uncertain domains

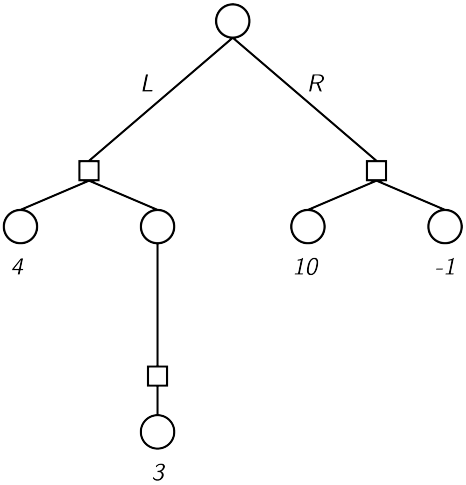
2 Application: Improving decisions with probabilistic reasoning

3 Bayesian networks

Plan

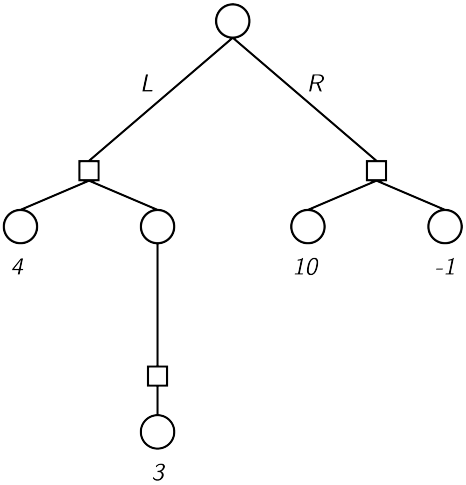
1. Uncertain domains
   - Random environments
   - Belief management
   - Basic properties of probabilities
   - Bayes rule
2. Application: Improving decisions with probabilistic reasoning
3. Bayesian networks

Why we need uncertainty: Back to the AND-OR trees



Optimal choice: The agent plays Left: when playing Right, there is a chance to obtain a very low outcome.

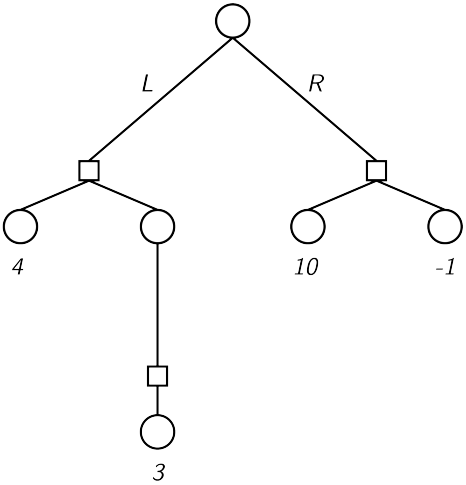
Why we need uncertainty: Back to the AND-OR trees



Optimal choice: The agent plays Left: when playing Right, there is a chance to obtain a very low outcome.

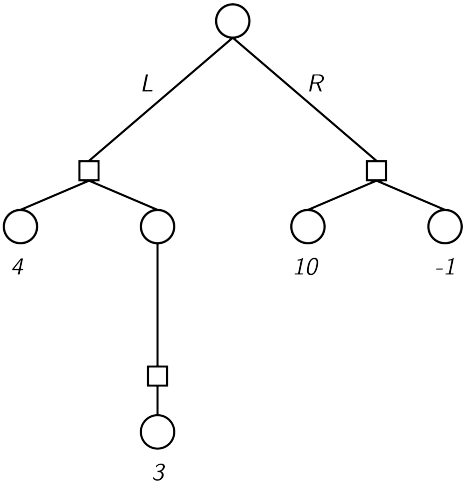
This example involves a random element (the OR nodes). But the randomness here is not *quantified*, so we need to consider the worst case scenario.

Why we need uncertainty: Back to the AND-OR trees



Now: We know that, if playing right, we obtain outcome 10 with probability 0.99 and -1 with probability 0.01.

Why we need uncertainty: Back to the AND-OR trees



Now: We know that, if playing right, we obtain outcome 10 with probability 0.99 and -1 with probability 0.01.

Optimal choice: Now, we know that, most of the times, we will win more by playing Right. This is a risk we can consider taking.

A first source of uncertainty: Random events

In this scenario, the environment is not deterministic: random events can happen.

The probability of an event corresponds to the **frequency** at which this event occurs.

Example: In the previous case, obtaining outcome 10 with probability 0.99 means that, if we play this situation 10,000 times, one might expect to get outcome 10 around 9,900 times.

A first source of uncertainty: Random events

In this scenario, the environment is not deterministic: random events can happen.

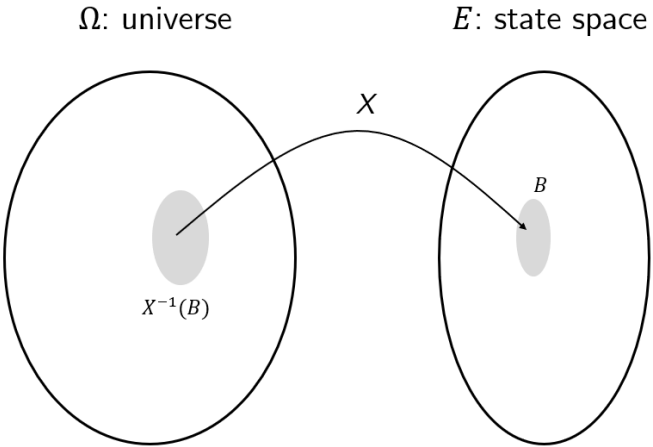
The probability of an event corresponds to the **frequency** at which this event occurs.

Example: In the previous case, obtaining outcome 10 with probability 0.99 means that, if we play this situation 10,000 times, one might expect to get outcome 10 around 9,900 times.

The **expected value** of some quantity X, denoted $\mathbb{E}[X]$, is the sum of values X can take, weighted by their probabilities.
It can be interpreted as the average value of X if the observation is repeated infinitely many times:

$$\mathbb{E}[X] = \int_x x \; dp(x) \simeq \frac{1}{N} \sum_{i=1}^{N} x_i$$

Reminder: Random variables and probabilities



Reminder: Random variables and probabilities

The **universe** $(\Omega, \mathcal{F}, \mathbb{P})$ is a space on which it is *easy* to *measure*. An element $\omega \in \Omega$ is a possible (but not directly observable) world.

The **state space** $(E, \mathcal{E})$ is a space of outcomes of a specific event (e.g. outcomes, actions...)

A **random variable** X is a function $X : \Omega \to E$ mapping an event in E to its corresponding possible worlds $\omega \in \Omega$, such that the image of "reasonable"[1] subsets of E are "reasonable" subsets of Ω.

We can then define the **probability distribution** $\mathbb{P}_X$ of the random variable X as the function mapping a "reasonable" subset B of events to:

$$\mathbb{P}_X(B) = \mathbb{P}(X^{-1}(B)) = \mathbb{P}(\{\omega \in \Omega : X(\omega) \in B\})$$

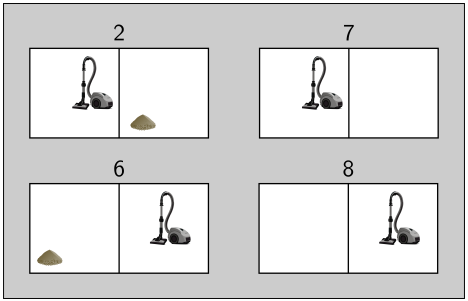
[1] "Reasonable" in the sense that it makes sense to *measure* them.

Plan

1. Uncertain domains
   - Random environments
   - Belief management
   - Basic properties of probabilities
   - Bayes rule
2. Application: Improving decisions with probabilistic reasoning
3. Bayesian networks

Why we need uncertainty: Back to the belief states

For the vacuum cleaner example with partial observability, observing no dust in the initial cell, we have the following initial belief state:



Question: How is this affected when we know that the environment is an old dusty house?

A second source of uncertainty: Uncertain beliefs

In this scenario, the agent gives weights to the states in the beliefs states: some seem more probable than some others.

The probability of a state corresponds to the **degree of belief** the agent has in the state.

In this case, the frequentist interpretation does not apply: probabilities do not interpret as the frequency to which a state is observed.

A second source of uncertainty: Uncertain beliefs

In this scenario, the agent gives weights to the states in the beliefs states: some seem more probable than some others.

The probability of a state corresponds to the **degree of belief** the agent has in the state.

In this case, the frequentist interpretation does not apply: probabilities do not interpret as the frequency to which a state is observed.

Question: Are these two visions of uncertainty coherent?

Plan

1. Uncertain domains
   - Random environments
   - Belief management
   - Basic properties of probabilities
   - Bayes rule
2. Application: Improving decisions with probabilistic reasoning
3. Bayesian networks

Conditional probabilities

The probability of event A corresponds to the measure of the worlds that make A possible.

Example: Probability that it is raining.

Conditional probabilities

The probability of event A corresponds to the measure of the worlds that make A possible.

Example: Probability that it is raining.

Sometimes, we have access to some extra information B, and want to know the probability of A given that B is true.

Example: Probability that it is raining knowing that it is sunny.

Such a probability is called the probability of A **conditionally** to B. This probability is equal to:

$$P_X(A|X) = \frac{P_X(A \wedge B)}{P_X(B)}$$

Joint distribution

We can group several random variables $X_1, \ldots, X_n$ into a tuple $\mathbf{X} = (X_1, \ldots, X_n)$, which is then a random variable. The distribution of $\mathbf{X}$ writes $P_{X_1,\ldots,X_n}(X_1, \ldots, X_n)$ and is called **joint distribution** of $X_1, \ldots, X_n$.

For any two sets of variables $\mathbf{X}$ and $\mathbf{Y}$, we have:

$$P(\mathbf{X}, \mathbf{Y}) = P(\mathbf{X}|\mathbf{Y})P(\mathbf{Y})$$

Remark: In the last equation, and in general, we omit to indicate as index the random variable of interest. This can be dangerous though...

Independent variables

Some variables seem completely **independent** from each other: For instance, $A =$ "Bob plays saxophone" and $B =$ "It is raining today".

When A and B are independent, we expect $P(A|B) = P(A)$. This is consistent with the definition of **independent variables**:

$$P(A, B) = P(A)P(B)$$

Remark: In general, when the variables are not independent, we have

$$P(A, B) = P(A|B)P(B)$$

Marginalization and conditioning

Consider two random variables X and Y.

Marginalization:

$$P(X) = \sum_{y} P(X, Y = y)$$

Marginalization and conditioning

Consider two random variables X and Y.

Marginalization:

$$P(X) = \sum_y P(X, Y = y)$$

Using the identity $P(X, Y) = P(X|Y)P(Y)$, we have:

Conditioning:

$$P(X) = \sum_y P(X|Y = y)P(Y = y).$$

Marginalization and conditioning: Examples

Example: What is the probability that Alice takes her umbrella when going out?

We choose the following notations:

- X indicates whether Alice takes her umbrella ($1 =$ she takes it, $0 =$ she doesn't)
- Y is the weather: can be *sunny* or *cloudy*.

We compute $P(X)$ by defining:

- The probability of the weather: *belief, can be obtained in a frequentist way*

$$P(Y = sunny) = 0.3 \quad \text{and} \quad P(Y = cloudy) = 0.7$$

- The probability of Alice's action: *usually frequentist*

$$P(X = 1|Y = sunny) = 0.1 \quad \text{and} \quad P(X = 1|Y = cloudy) = 0.8$$

Marginalization and conditioning: Examples

Example: What is the probability that I hear people speaking Norwegian in the street?

We choose the following notations:

- X indicates whether I hear some Norwegian ($1 =$ I do, $0 =$ I don't)
- Y is the country I am in

We compute $P(X)$ by defining:

- The probability of the country: *belief only, based on my knowledge of where I am*
- The probability of hearing Norwegian: *frequentist*

$$P(X = 1|Y = Norway) = 0.85, \quad P(X = 1|Y = Sweden) = 0.1, \quad \ldots$$

Plan

1. Uncertain domains
   - Random environments
   - Belief management
   - Basic properties of probabilities
   - Bayes rule
2. Application: Improving decisions with probabilistic reasoning
3. Bayesian networks

A simple observation

In joint distribution, the order of variables does not matter:

$$P(X, Y) = P(Y, X)$$

As a consequence, we can propose the following identity:

$$P(X|Y)P(Y) = P(Y|X)P(X)$$

the rewriting of which is known as the **Bayes rule**:

$$P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)}$$

Example

Suppose that I want to guess whether I am currently in Norway.

I am originally not sure of where I am, and I evaluate my chances to be in Norway to 0.3. This is called the **prior**: belief about my state prior to observing anything.

Example

Suppose that I want to guess whether I am currently in Norway.

I am originally not sure of where I am, and I evaluate my chances to be in Norway to 0.3. This is called the **prior**: belief about my state prior to observing anything.

I know that, if I am in Norway, I have probability 0.85 to hear Norwegian, and 0.05 if I am not. This is called the **likelihood**: how likely it is to make some observations knowing the state.

Example

Suppose that I want to guess whether I am currently in Norway.

I am originally not sure of where I am, and I evaluate my chances to be in Norway to 0.3. This is called the **prior**: belief about my state prior to observing anything.

I know that, if I am in Norway, I have probability 0.85 to hear Norwegian, and 0.05 if I am not. This is called the **likelihood**: how likely it is to make some observations knowing the state.

The probability that I am in Norway knowing my observation is called **posterior probability**.

Example

For instance, if I heard some people speaking Norwegian outside ($X = 1$), I have:

$$P(Y = Norway|X = 1) = \alpha P(X = 1|Y = Norway)P(Y = Norway) = 0.85 \times 0.3 \times \alpha = 0.255\alpha$$
$$P(Y \neq Norway|X = 1) = \alpha P(X = 1|Y \neq Norway)P(Y \neq Norway) = 0.05 \times 0.7 \times \alpha = 0.035\alpha$$

α is a normalization factor: $P(Y = Norway|X = 1) + P(Y \neq Norway|X = 1) = 1$, so $0.29\alpha = 1$ i.e. $\alpha \simeq 3.45$

In the end, we have that, after observation, the probability of being in Norway is

$$P(Y = Norway|X = 1) \simeq 0.88$$

Outline

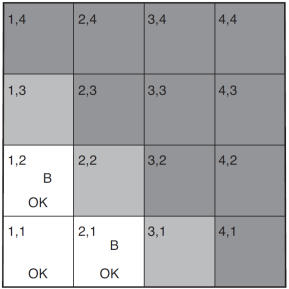
1 Uncertain domains

2 Application: Improving decisions with probabilistic reasoning

3 Bayesian networks

Our problem: The Wumpus world

In the Wumpus world, we might end up in a case where there is no safe cell to explore.



Logic does not allow to state which cells are safer to explore next. Let us use probabilities to determine this.

Task description

We use the same formalism as in description logic:

- P_{ij} means that there is a pit in cell (i, j)
- B_{ij} means that there is a breeze in cell (i, j)

Task description

We use the same formalism as in description logic:

- P_{ij} means that there is a pit in cell (i, j)
- B_{ij} means that there is a breeze in cell (i, j)

We assume that a cell can contain a pit with probability 0.2 (which corresponds to a **prior belief**), and that the positions of pits from on cell to the other are independent:

$$P(P_{11}, P_{12}, \ldots, P_{44}) = \prod_{1 \leq i,j \leq 4} P(P_{ij})$$

Task description

We use the same formalism as in description logic:

- P_{ij} means that there is a pit in cell (i, j)
- B_{ij} means that there is a breeze in cell (i, j)

We assume that a cell can contain a pit with probability 0.2 (which corresponds to a **prior belief**), and that the positions of pits from on cell to the other are independent:

$$P(P_{11}, P_{12}, \ldots, P_{44}) = \prod_{1 \leq i,j \leq 4} P(P_{ij})$$

Full joint distribution of the grid:

$$P(P_{11}, B_{11}, \ldots, P_{44}, B_{44}) = P(B_{11}, \ldots, B_{44} | P_{11}, \ldots P_{44}) P(P_{11}, \ldots, P_{44})$$

Our problem

We want to estimate the posterior probability $P(P_{13}|Observations)$, where the observations consist of:

- Observation of the cells with no pits: $p_{obs} = (\neg p_{11}, \neg p_{12}, \neg p_{21})$
- Observation of the breeze in the cells: $b_{obs} = (\neg b_{11}, b_{12}, b_{21})$

Question: Can we compute $P(P_{13}|p_{obs}, b_{obs})$?

Simplification based on normalization

We notice that:

$$P(P_{13}|p_{obs}, b_{obs}) = \frac{P(P_{13}, p_{obs}, b_{obs})}{P(p_{obs}, b_{obs})}$$

Since p_{obs} and b_{obs} are observed, they are constants. Consequently, we can write:

$$P(P_{13}|p_{obs}, b_{obs}) = \alpha P(P_{13}, p_{obs}, b_{obs})$$

Simplification based on normalization

We notice that:

$$P(P_{13}|p_{obs}, b_{obs}) = \frac{P(P_{13}, p_{obs}, b_{obs})}{P(p_{obs}, b_{obs})}$$

Since p_{obs} and b_{obs} are observed, they are constants. Consequently, we can write:

$$P(P_{13}|p_{obs}, b_{obs}) = \alpha P(P_{13}, p_{obs}, b_{obs})$$

Remark: In practice, α can be obtained by noticing that

$$P(P_{13}|p_{obs}, b_{obs}) + P(\neg P_{13}|p_{obs}, b_{obs}) = 1$$

Further simplifications

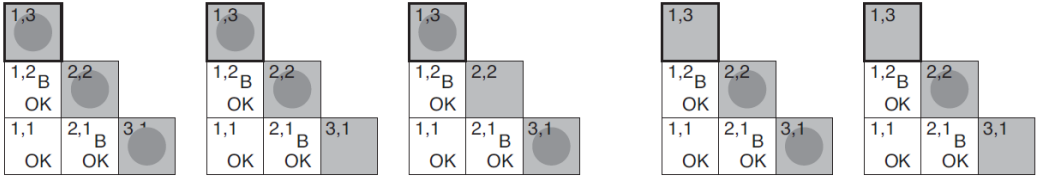
$$
\begin{aligned}
P(P_{13}|p_{obs}, b_{obs}) &= \alpha P(P_{13}, p_{obs}, b_{obs}) \\
&= \alpha \sum_{p_{nobs}} P(P_{13}, p_{obs}, b_{obs}, p_{nobs}) \\
&= \alpha \sum_{p_{nobs}} P(b_{obs}|P_{13}, p_{obs}, p_{nobs}) P(P_{13}, p_{obs}, p_{nobs}) \\
&= \alpha \sum_{p_{frontier}} \sum_{p_{others}} P(b_{obs}|P_{13}, p_{obs}, p_{frontier}, p_{others}) P(P_{13}, p_{obs}, p_{frontier}, p_{others}) \\
&= \alpha \sum_{p_{frontier}} \sum_{p_{others}} P(b_{obs}|P_{13}, p_{obs}, p_{frontier}) P(P_{13}, p_{obs}, p_{frontier}, p_{others}) \\
&= \alpha \sum_{p_{frontier}} P(b_{obs}|P_{13}, p_{obs}, p_{frontier}) \sum_{p_{others}} P(P_{13}) P(p_{obs}) P(p_{frontier}) P(p_{others})
\end{aligned}
$$

Further simplifications (continued)

$$
\begin{aligned}
P(P_{13}|p_{obs}, b_{obs}) &= \alpha \sum_{p_{frontier}} P(b_{obs}|P_{13}, p_{obs}, p_{frontier}) \sum_{p_{others}} P(P_{13})P(p_{obs})P(p_{frontier})P(p_{others}) \\
&= \underbrace{\alpha P(p_{obs})}_{=\alpha'} P(P_{13}) \sum_{p_{frontier}} P(b_{obs}|P_{13}, p_{obs}, p_{frontier})P(p_{frontier}) \underbrace{\sum_{p_{others}} P(p_{others})}_{=1} \\
&= \alpha' P(P_{13}) \sum_{p_{frontier}} P(b_{obs}|P_{13}, p_{obs}, p_{frontier})P(p_{frontier})
\end{aligned}
$$

Remark: $P(b_{obs}|P_{13}, p_{obs}, p_{frontier})$ can be equal only to 1 (in case the frontier is compatible with the observed breezes) or 0 (in case it is not)

Possible frontiers



$$P(P_{13}|p_{obs}, b_{obs}) = 0.2 \times (0.04 + 0.16 + 0.16) \times \alpha' = 0.072\alpha'$$
$$P(\neg P_{13}|p_{obs}, b_{obs}) = 0.8 \times (0.04 + 0.16) \times \alpha' = 0.16\alpha'$$

which gives in particular $0.232\alpha' = 1$, so $\alpha' \simeq 4.31$

Comparing the results

We compare the probabilities obtained for each of the possible cells to explore:

$$P(P_{13}|p_{obs}, b_{obs}) \simeq 0.31$$
$$P(P_{22}|p_{obs}, b_{obs}) \simeq 0.86$$
$$P(P_{31}|p_{obs}, b_{obs}) \simeq 0.31$$

It is then clear that the agent should avoid the cell $(2, 2)$. However, this is just a probabilistic reasoning: there is no guarantee that the pit is not in $(1, 3)$ or $(3, 1)$!

Outline

1 Uncertain domains

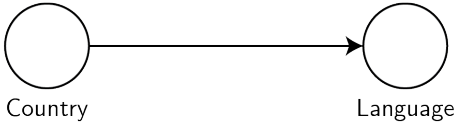
2 Application: Improving decisions with probabilistic reasoning

3 Bayesian networks

Back to the example of country identification

Example: We want to identify the fact that we may be in Norway, depending on which language we hear in the street.

In this example, we have 2 variables (country, language heard), linked by a conditional probability distribution. We can represent this graphically:



In this case we have:

$$P(C, L) = P(C)P(L|C)$$

Bayesian Networks

A **Bayesian network** is a probabilistic model representing joint distributions of variables in the form of a directed acyclic graph (DAG):

- The nodes of the graph correspond to random variables of interest
- Edges represent conditional distribution between variables

$$P(X_i|Parents(X_i))$$

Bayesian Networks

A **Bayesian network** is a probabilistic model representing joint distributions of variables in the form of a directed acyclic graph (DAG):

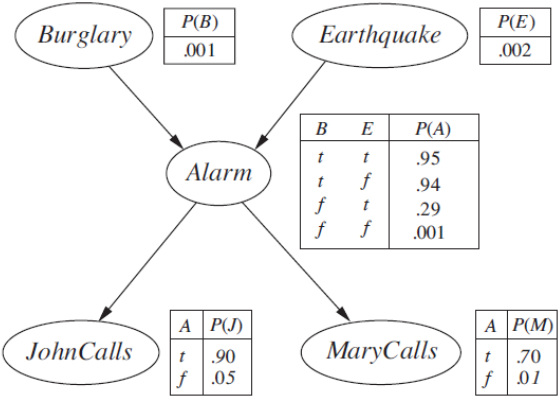
- The nodes of the graph correspond to random variables of interest
- Edges represent conditional distribution between variables

$$P(X_i|Parents(X_i))$$

The full joint probability over all variables $(X_1, \ldots, X_n)$ is given by:

$$P(X_1, \ldots, X_n) = \prod_{i=1}^{n} P(X_i|Parents(X_i))$$

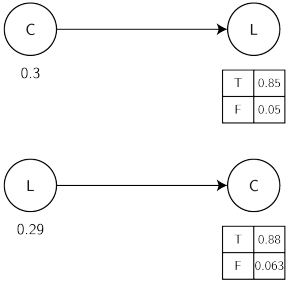
An example of a Bayesian network



Bayesian Networks and Causality

It is tempting to interpret Bayesian Networks as **Causal networks**: an edge from X_i to X_j would mean that X_i causes X_j.

However, this is incorrect: from a global point of view, it is possible to reverse all edges and keep the same joint distribution.

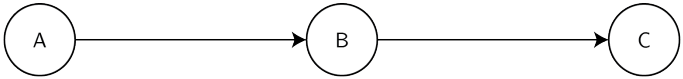


Variable independence

The key property of BNs introduce a dependence between each variable and its parents:

$$P(X_i|Parents(X_i))$$

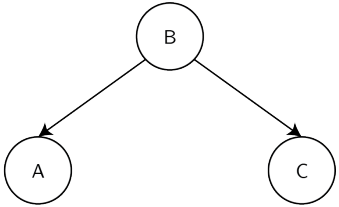
This does **not** mean that there is independence to all variables that are not the parents!



In this example, C and A are obviously dependent. However, if given B, C is independent of A. This is called **conditional independence**.

$$P(C|A,B) = \frac{P(A,B,C)}{P(A,B)} = \frac{P(A)P(B|A)P(C|B)}{P(A)P(B|A)} = P(C|B)$$

Variable independence: Common cause



In this example, C and A are dependent because they share a common cause. But they are independent conditionally to the cause:

$$P(C|A,B) = \frac{P(A,B,C)}{P(A,B)} = \frac{P(B)P(A|B)P(C|B)}{P(B)P(A|B)} = P(C|A,B)$$

Inference in Bayesian Networks

An essential task to perform in a Bayesian Network is **inference**: given some observe variables $E_1, \ldots, E_k$, called **evidence** and a query variable Q, compute

$$P(Q|E_1, \ldots, E_k)$$

In principle, this is easy:

$$P(Q|E_1, \ldots, E_k) = \frac{P(Q, E_1, \ldots, E_k)}{\sum_q P(q, E_1, \ldots, E_k)} = \frac{\sum_{h_1,\ldots,h_r} P(Q, E_1, \ldots, E_k, h_1, \ldots, h_r)}{\sum_q P(q, E_1, \ldots, E_k)}$$

where $h_1, \ldots, h_r$ are the values that each unobserved (= **hidden**) variable H_i can take.

Inference in Bayesian Networks

An essential task to perform in a Bayesian Network is **inference**: given some observe variables $E_1, \ldots, E_k$, called **evidence** and a query variable Q, compute

$$P(Q|E_1, \ldots, E_k)$$

In principle, this is easy:

$$P(Q|E_1, \ldots, E_k) = \frac{P(Q, E_1, \ldots, E_k)}{\sum_q P(q, E_1, \ldots, E_k)} = \frac{\sum_{h_1,\ldots,h_r} P(Q, E_1, \ldots, E_k, h_1, \ldots, h_r)}{\sum_q P(q, E_1, \ldots, E_k)}$$

where $h_1, \ldots, h_r$ are the values that each unobserved (= **hidden**) variable H_i can take.

This method is called **inference by enumeration** but is not tractable in general.

An example of approximate inference: Rejection sampling

Sampling consists in simulating the outcome of a probability distribution.

In a Bayesian Network, sampling can be done recursively from the root nodes.

An example of approximate inference: Rejection sampling

Sampling consists in simulating the outcome of a probability distribution.

In a Bayesian Network, sampling can be done recursively from the root nodes.

In **rejection sampling**, the idea is to estimate $P(Q|E_1, \ldots, E_k)$ by:

- Sampling from the BN full distribution
- Eliminating all samples which are not consistent with $E_1, \ldots, E_k$
- Observe the outcome Q in the kept samples.

An example of approximate inference: Rejection sampling

Sampling consists in simulating the outcome of a probability distribution.

In a Bayesian Network, sampling can be done recursively from the root nodes.

In **rejection sampling**, the idea is to estimate $P(Q|E_1, \ldots, E_k)$ by:

- Sampling from the BN full distribution
- Eliminating all samples which are not consistent with $E_1, \ldots, E_k$
- Observe the outcome Q in the kept samples.

Problem: When the rejection conditions are too restrictive, one might have to sample an exponentially large number of points to have reasonable accuracy. Other techniques are needed...

Other inference techniques?

Rejection sampling is a simple **approximate inference** method, but it does not scale well...
What could be other inference strategies?

Other inference techniques?

Rejection sampling is a simple **approximate inference** method, but it does not scale well...
What could be other inference strategies?

Answer in winter semester, in the course: "Probabilistic Machine Learning"!